

DEXTER FERNANDES

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SUMMARY

Computer vision and perception engineer with an MSc in Robotics (Merit) and 3+ years of commercial computer vision experience. Built **LiDAR SLAM** systems in C++17 and **ROS2** on a Clearpath Husky, applying **sensor fusion** across LiDAR, depth camera and IMU data and **point cloud** processing with PCL and Open3D. Shipped real-time object detection to industrial sites using classical and learning-based methods, debugging perception system behaviour directly from raw sensor data in messy, real-world conditions.

RELEVANT EXPERIENCE

Esbaar — AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Owned the end-to-end ML lifecycle for real-time perception on BP rigs, from data curation and training through deployment and on-site validation, improving F1 by 15% under 24/7 operation.
- Adapted YOLOv5 **object detection** models in PyTorch, replacing backbone and neck layers and retraining for rig-specific classes, with Optuna sweeps and **Weights & Biases** tracking across runs.
- Ran TIDE error analysis on COCO-format production data to split live detector failures by error type, directing dataset fixes and threshold tuning at the dominant one.
- Developed and maintained NVIDIA DeepStream and FFmpeg video pipelines for 24/7 perception on BP drilling rigs, covering camera ingestion, inference and field diagnostics in desert conditions.
- Led on-site validation on BP rig deployments, diagnosing model failures under lighting, motion and environmental stressors and turning findings into fixes for live operation.
- Built ingestion and validation pipelines across image, video and sensor data with Airflow, **MLflow**, DVC and CVAT, processing 3.6 million records and cutting rollout from 72h to under 24h.
- Applied image preprocessing and artefact detection with OpenCV and scikit-image to catch mud and sand contamination on camera feeds, reducing false positives by 13%.
- Built a watchdog application that detected and restarted failed services on edge devices, and designed its REST API exposing service health and restart control, removing the need for site visits to recover pipelines.

LivNSense Technologies — Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Led four engineers building a two-stage detector feeding helmet-colour classification for construction safety in Texas, cutting false positives by 33% within latency and power budgets.
- Built asynchronous **C++17** perception pipelines on OpenVINO across Intel Movidius VPUs and Core i7 industrial edge boxes, raising throughput from 5 to 15 FPS under real-time constraints.
- Built a real-time multi-camera perception pipeline on a moving construction vehicle, synchronising streams and running incident detection under edge compute constraints.
- Added ByteTrack multi-object tracking over detections to hold worker and vehicle identities across frames, cutting duplicate incident alerts from the moving-vehicle feeds.
- Optimised architectures with quantisation-aware training, structured pruning and layer fusion in Keras and OpenVINO, cutting model size by 50% while holding production accuracy.
- Used confusion-matrix failure clustering and TIDE error analysis on COCO-format production data to diagnose recurring detector failures and prioritise dataset fixes across camera feeds.
- Automated labelling by repurposing existing detection models and added dataset versioning with DVC, cutting manual labelling effort by 80% on large real-world datasets.
- Evaluated DeepLabv3+, Mask R-CNN, U-Net and SegFormer for annotation review, foreground separation and failure analysis on industrial camera feeds.

PROJECTS

ROS2 SLAM and Nav2 Simulation

02/2026

- Built a simulation-based robotics stack in **ROS2** Jazzy and Gazebo Harmonic to evaluate **LiDAR**-centric SLAM, LiDAR and RGB fusion SLAM, and autonomous navigation in structured scenarios.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer and GLIM across 2D and 3D LiDAR configurations, comparing mapping and localisation behaviour to support system selection.
- Implemented **point cloud** filtering, segmentation and registration with PCL and **Open3D**, and added an EKF localisation layer with robot_localization fusing wheel odometry and IMU.
- Developed ROS2 nodes, launch files and parameterised configs in **C++** and **Python**, using tf2, message synchronisation and timestamp alignment to debug transform timing issues.

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Built an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline running YOLO detection and NvDCF tracking across 3 concurrent feeds, with metadata extraction, anonymisation and re-streamed output.
- Converted YOLO26n to a dynamic-batch TensorRT FP16 engine on 6GB VRAM Turing hardware, validating FP16 against FP32 accuracy and profiling p99 latency to sustain 29.7 FPS per stream live.
- Benchmarked IoU, NvDCF and NvSORT trackers on MOTA, IDF1, ID switches, fragmentation, FPS and VRAM impact to guide tracker selection.

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking

03/2026

- Benchmarked modern PyTorch detectors including YOLOv8, YOLOv10, YOLO26 and RT-DETR on accuracy, latency and model size to compare deployment suitability for edge targets.
- Extended the comparison to EfficientViT, MobileSAM and Depth Anything v2 to assess whether segmentation and depth models are viable under the same edge constraints.
- Evaluated Vision Transformer and foundation-model approaches across detection, segmentation, depth estimation and open-vocabulary understanding.

EDUCATION

University of Bristol — MSc Robotics, Merit 09/2023 – 09/2024 · BRISTOL, ENGLAND

Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time **SLAM** system in C++17 and ROS2 (roslcpp) on a Clearpath Husky, fusing Velodyne VLP-16 **LiDAR**, RealSense D435i depth and radiological sensing for inspection mapping.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched rosbag2 sequences under a single evaluation protocol to compare mapping and localisation behaviour.
- Configured Ceres, g2o and GTSAM as solver backends and tuned ICP parameters across environments.
- Calibrated LiDAR extrinsics to the robot base and resolved transform timing with tf2 across a **PCL point cloud** pipeline, validating sensor placement for coverage.

Visvesvaraya National Institute of Technology — BTech Electrical and Electronics Engineering 07/2016 – 05/2020 · NAGPUR, INDIA

- Foundation in signals, systems, embedded hardware and control, which later supported work in real-time AI, sensor systems and robotics software.

SKILLS

Languages	Python, C++17, CUDA C++, Bash, SQL
Robotics and SLAM	ROS2 (Robot Operating System 2), roslcpp, roslpy, tf2, rosbag2, Gazebo, RTAB-Map, LIO-SAM, Cartographer, GLIM, Nav2, LiDAR SLAM, pose graph optimisation, ICP (Iterative Closest Point), scan matching, LiDAR extrinsics calibration, PCL (Point Cloud Library), Open3D, Eigen, Ceres, g2o, GTSAM, robot_localization, EKF
Computer vision and DL	object detection, image classification, semantic and instance segmentation, object tracking, optical flow, data augmentation, CNNs, YOLO (v5/v8/v10/v26), RT-DETR, EfficientViT, MobileSAM, Depth Anything v2, PyTorch, TensorFlow, Keras, OpenCV, scikit-image, NumPy
Edge inference	TensorRT, OpenVINO, ONNX, quantisation-aware training, post-training quantisation, FP16/INT8, structured pruning, latency and throughput profiling, NVIDIA Jetson AGX Xavier, Intel Movidius Myriad VPU
Video pipelines	NVIDIA DeepStream, GStreamer, FFmpeg, RTSP, multi-stream video analytics, multi-camera synchronisation, ByteTrack, NvDCF, NvSORT
MLOps and experiments	MLflow, DVC (Data Version Control), Apache Airflow, Optuna, Weights & Biases, Jupyter, CVAT (Computer Vision Annotation Tool), dataset versioning, synthetic data generation, benchmarking harnesses, ablation studies, failure-mode analysis, TIDE
Production and cloud	Docker, CI/CD, Git, CMake, pytest, unittest, gtest, REST API design, containerised microservices, multiprocessing, Linux
Time series and ML	gradient boosting, random forests, scikit-learn, Pandas, multivariate time-series classification
Professional	technical leadership, mentoring, Agile and Scrum, stakeholder communication, cross-functional delivery

ADDITIONAL EXPERIENCE

GrowthStage — Technical Assessor, Computer Vision and Software Engineering 05/2026 – PRESENT · REMOTE

Assesses candidate technical depth across perception, SLAM and deployment using hands-on robotics and computer vision knowledge.

Taco Bell — Service Champion / Team Trainer 09/2024 – PRESENT · BRISTOL, ENGLAND

Progressed to Team Trainer in a customer-facing environment, onboarding new staff and handling escalations.

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019) Best Speaker award, ABC Toastmasters Club (01/2020)